The Selection and Management of Contractors

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ABSTRACT

The mining industry in Australia frequently uses the services of specialist engineering and managing contractors and consultants for a majority of the work required for project development, from feasibility studies to project commissioning. The project owner's prime objectives are to achieve predictability and competitive advantage for their project. As the majority of this work will be carried out by contractors, the effective evaluation, selection and management of these contractors in the most appropriate forms of contracts is an essential ingredient to project success.

The first stage to project development should be for the owner to examine its own resources to establish the role that it will fill and thereby identify the role and scope of contractors' involvement.

Once the owner has developed this work plan, the selection of contractors and consultants should be structured to ensure that the owner acquires the appropriate expertise and experience to achieve its project objectives. The sequence of selection procedures is:

1. Pre-qualification of prospective contractors according to the owner's adopted criteria, through the use of a pre-qualification questionnaire.
2. Evaluation by the owner tender board according to predetermined mandatory criteria.
3. Preparation of tender documents for submission to qualified contractors.
4. Evaluation of tender submissions by the owner tender board using a pre-determined, weighted matrix of technical and commercial criteria.
5. Final selection and contract negotiation.

In addition, the owner's work plan should identify the most appropriate form of contract for each phase of project development. This will be based on the degree of project definition and the owner's requirements for the development schedule, risk structure and commercial status of the project.

The owner's role in management of the work will vary according to the form of contract adopted. The flow of timely and appropriate information between the participants in project development will be ensured by the preparation of a Project Procedures Manual at the time of commitment.

INTRODUCTION

The prime objectives of the owner of a project are to achieve:

1. Competitive advantage: best, low cost, most profitable project.
2. Predictability: what is forecast shall actually occur.

A recent study by the Victorian Department of Industry, Technology and Resources (DITR) indicated deficiencies in the predictability of Australian project performance against time and cost criteria, when compared to other countries. Another study, by Mulligan and Williams-Wynn, into the Petrochemical Industry, concludes that Australia suffers a considerable cost disadvantage, in terms of capital inputs, compared to Australia's major competitors in that industry. The mining industry may not necessarily suffer from the same cost disadvantages as the petrochemical industry, but the engineering of the best, lowest cost project, the precise estimate of the costs to develop that project and subsequent management to achieve the project objectives are as critical to the mining industry as to any other industry. The DITR report indicates that the mining industry is very much more successful than public service industries in achieving time and cost objectives. It also mentions that managing contractors are more successful than owner managers and that inaccurate cost estimating and budgeting is the most frequent reason for projects not performing to targets.

A paper by Frank Gulliver provided four guidelines to project success:

1. Determine costs accurately.
2. Anticipate and minimise risk.
3. Evaluate contractors.
4. Improve project management.

While the other papers at this symposium are primarily concerned with the first of these guidelines, this paper touches on the other three and, in particular, addresses the evaluation of contractors. Generally, few owner organisations have sufficient resources to carry out all the technical and management functions required for feasibility studies and project implementation and, as an industry, we rely heavily on managing contractors. Thus the selection and management of these contractors is a critical step in the process of achieving project objectives and ensuring that cost estimates are accurately prepared to provide the predictability requirements of the owner.

The objective of this paper is to present an owner perspective on the owner-managing contractor process with a view to more predictable, cost effective projects.

OWNER ROLE

Typically a project will progress through a series of studies of increasing definition, complexity and precision of cost estimating, prior to a decision being made on implementation. At some stage in this process, it is assumed that the owner will require the assistance of contracting organisations, in particular to develop engineering plans and capital cost estimates.

It is up to the owner to define what will be the role of the contractors and at what stage their input will replace or enhance that of owner's staff. The small entrepreneurial company will rely more heavily on contractors than major companies that are well established in the mining industry. Typically though the role of feasibility study preparation and the subsequent management of project implementation are contracted to managing contractors or are divided between a series of engineering contractors with expertise in specific disciplines.

Hence the first step for the owner is to evaluate its own resources, in relation to the particular project, to establish what and who is required in the way of external assistance. It is also up to the owner to decide what it believes will be the final

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product of that assistance; that is, to define the scope of the contractors' work. If, for example, the final product of that phase of the project is to be a bankable feasibility document and the project will employ a processing technology for which the owner cannot supply design criteria, then the requirements for contracting assistance are markedly different to those of an owner who can supply all process design criteria.

This paper advocates a formal and structured approach to the identification, selection and management of contractors. By the owner being specific in the definition of the role of contractors, disputes and misunderstandings will be minimised, which will be to the overall benefit of both the owners' and the contractors' profitability.

PRE-QUALIFICATION

Listing of Potential contractors

Once the owner organisation has identified the requirements for work by contractors in relation to the project, the next stage is to identify and draw up a list of contracting organisations that may qualify for this work. The first listing should be broad based, as the objective of the pre-qualification phase is to define the qualities and assets of those contractors with whom it is worth proceeding to formal tendering for further work. Being broad based, the names of potential contracting organisations may be sourced from the technical literature or from industry contacts. There is little concern that all interested parties will be involved, as the industry grapevine usually puts any omissions from the list in touch with the owner. At this stage the owner has no real reason to be overly selective as to who to include on the list. Selection should only commence after a response to the pre-qualifications questionnaire.

Pre-Qualification Questionnaire

The pre-qualification questionnaire should be structured according to the owner's requirement for further information from companies with whom, at some stage, a contract may be entered into. The work content and scope of the intended contracts will influence the content and detail required by the owner. A questionnaire to ensure the integrity of a consultant for a single task specific to that consultant's expertise will be less comprehensive than that required from contractors who may be lump sum constructing a major item of work. However, it is considered that the principles of the questionnaire are generally the same in both cases.

The contents presented below are suggested where comprehensive information is required. This form of questionnaire is based on the FIDIC Standard Pre-Qualification form for contractors.

The questionnaire provides the contracting organisations with all necessary information by which they will be able to assess their interest in proceeding further, and then to extract information from the contractors that will be pertinent to the owner's evaluation and selection process. An unstructured approach may result in the receipt of standard brochures and experience listings, while the questionnaire, though still providing these items, will provide more definitive and specific information by which to judge the relative competency of the potential tenderers.

The questionnaire covers three areas:

1. Terms and conditions for future business.
2. Information supplied to the contractor.
3. Information required from the contractor.

TABLE 1

Questionnaire to qualify for bid list status

Table of Contents

Introductory Letter

1. Non-disclosure letter agreement
2. Scope of Work
3. Project Information
   - 3.1 Ownership and Management
   - 3.2 Project Location
   - 3.3 Project Description
   - 3.4 Special Considerations
4. Pre-Qualification Questionnaire
   - 4.1 Company Structure
   - 4.2 Management Structure/Capabilities
   - 4.3 Financial Information
   - 4.4 Resources - Staff and Equipment
   - 4.5 Activity Level
   - 4.6 Industrial Relations
   - 4.7 Safety
   - 4.8 Environmental
   - 4.9 Quality Management
   - 4.10 Project Control
   - 4.11 Sub-contractors and Consultants
   - 4.12 Experience
5. General Conditions of Contract

Terms and Conditions

The inclusion of a non-disclosure letter agreement assumes that the owner considers that their plans for the project, and any information that may be supplied in the future, is information proprietary to the owner. If this is the case, then it is as well to establish this at the outset and to formalise this situation by having interested parties sign to their agreement of non-disclosure of project information. The documents will also specify the conditions under which information should be returned to the owner. A further purpose of the non-disclosure agreement may be in relation to a process patented or licensed by one or other contractor. That contractor would be in a position to specify exclusion of the patented process from the owner's ownership of project information.

It is recommended that the pre-qualification questionnaire includes the owner's General Conditions of Contract, particularly if these are significantly different from A.S. 2124-1986. If A.S. 2124 is to be used, then this should be specified. Including owner's terms and conditions at the pre-qualification stage provides potential contractors the opportunity of taking exception to specific terms. These could be of sufficient importance to the owner that the exceptions could disqualify a contractor. It is obviously advantageous to disqualify at an early stage rather than to spend many hours of follow-up by both contractors and owner, only to discover later a contractual incompatibility. Examples of potential exceptions:

1. Independent contractor status is generally unacceptable to consulting engineers.
2. Professional indemnity insurance or liquidated damages provisions.
3. Parent company guarantees.

Project Information

Sections 2 and 3 of the questionnaire specify the owner's perception of what work is considered to be the contractor's scope. Sufficient information of the project and the owner is supplied to allow the contractor full opportunity to reply to the questionnaire in the most advantageous terms.

Special considerations, for example, environmental conditions, safety requirements, quality management standards, or the specific conditions of an overseas location should be stated, to ensure the quality of the contractors' replies.

Contractor Supplied Information

Major contracting organisations often assume responsibility for the effective use of considerable amounts of the owner's money. Under this circumstance, it is not unreasonable for an owner to want in-depth knowledge of that contractor. This will include:

1. company structure: ownership, joint venturers, organisation chart.
2. financial status: Dunn and Bradstreet report, annual report, liabilities, indemnity provisions.
3. resources and commitments: staff, equipment, jobs in progress/bidding, activity forecast.
4. expertise: industrial relations, safety, environment, quality management, project control, patents/licenses, sub-contractors and consultants.
5. experience: project list, references.

Most major contractors would have all of this information readily at hand and it has been remarked, though not by all, that the specific questionnaire is time saving to them in the preparation of their submission. A contractor may also appreciate the even-handedness of the questionnaire approach.

FORM OF CONTRACT

To return to the theme of predictability, most owners would wish to know how much a certain scope of work would cost and would then wish to enter into an agreement such that the work would cost just that, and not any more. This situation is only achievable if the scope of work is precisely defined and is not subsequently changed. A lump sum contract, as defined above, can readily be achieved and, in the case of a construction contract, generally requires that about 50% of the engineering definition is already completed. Therefore the concept of lump sum contracting is generally incompatible with:

1. feasibility studies
2. fast tracking
3. owner involvement

The owner should, in all cases, examine the most appropriate form of contract in relation to the proposed development schedule, contract value, risk structure, owner expertise and organisation and project financing arrangements. Most contracts fall into one of the following five categories:

1. Reimbursable Contract. owner pays for time and materials at agreed rates. Cost and schedule risks are totally to the owner. Generally, when many options remain, e.g. feasibility study, this is the most realistic form of contract.
2. Schedule of Rates. Rates for various activities are agreed to on the basis of a quantity survey from partially completed engineering. The owner bears the cost and schedule risks. For fast tracking construction works, mining contracts or for construction contracts that are so large that contractors will not assume the risk, this is the most appropriate type of contract.
3. Target (Bonus/Penalty). With a reasonable quality of project definition, the contractor can agree to targets for both development cost and schedule. The contract is structured such that the contractor takes a part of the benefit from cost or schedule underruns or progressively forfeits fee and reimbursable payments in the case of cost or schedule overruns. This partial sharing of risk and benefits creates a situation of shared objectives, to the potential advantage of both parties.
4. Lump Sum. An agreed work scope is implemented for an agreed fixed cost. Mechanisms are included for handling and costing variations of scope. The risk is for the contractor and, because of this, there are occasions when lump sum tender costs are padded to reduce the risk to the contractor. Therefore lump sum contracts, while they may provide predictability, are not necessarily the lowest cost means of implementing a scope of work. On large contracts, it may not be possible to obtain lump sum bids at all, because contractors are unwilling to carry the value of the risk.
5. Turnkey. Generally when a company or contractor is the provider of unique technology or know-how, an all up-price will be agreed to covering design, procurement, supply, construction and commissioning of the facilities covered by that technology.

It is worth emphasizing that the lump sum contract is not always the best solution. An example is a project in which the acceptance of lump sum tenders from either of two major contracting groups would have resulted in budget overruns of 13.5% and 18% respectively. The Target (Bonus/Penalty) route was selected by the owner, and the final cost underrun budget by 5%.

CONTRACTOR SELECTION

The process of contractor selection comprises:

1. Short-listing
2. Tendering
3. Tender Evaluation
4. Contract negotiation with the selected company

Short-listing

The recommended process for evaluating tender documents is to form a tender board or committee that will establish the premises for evaluation, prior to opening the tenders. A tender board of four persons, two technical, one commercial and one executive, is usually sufficient for all but the largest and most complex tender evaluations.

The short-listing process, based on the pre-qualification submission, should be to identify those companies that are judged to be able to handle the scope of work.

In order to formalise the evaluation process, the tender board should establish a matrix of conditions by which to judge the tenders. In the case of short-listing, each of these conditions will be mandatory; i.e. if a submission is judged to fail any condition, then that contractor will be eliminated.

As an example, the tender board may decide on the following "must" conditions as a pre-requisite to inclusion in a short list for engineering and construction work on a project in which the contractor would be expected to take a part of the financial risk.

1. financial stability and reputation
2. organisation capability
3. proven track record
4. current availability of resources

It is worth noting that the "must" conditions are entirely dependent upon the owner's project requirements and the work scope. For example, for feasibility work, the following conditions may be set:

1. technical expertise
2. proven track record
3. bankable document reputation
4. current availability of resources

These conditions reflect the nature of feasibility work, which is to produce technically viable concepts, that are then accurately estimated, and then to produce a comprehensive and legible feasibility document. The qualities are substantially different from those required for detailed design and construction, a consideration that should be kept in mind before committing to continue with a feasibility contractor into construction.

The tender board would judge each submission in committee in order to reach consensus on each condition, thereby providing "go" or "no-go" decisions on each.

Tendering

The tendering process is similar to that for the pre-qualification exercise, except that more detail is provided by the owner and is required from the contractor.

If the tendering is for construction and there is a feasibility study and back-up information, then it is advantageous to make all of this available in a project office, giving each of the short-listed tenderers one day to review the information. A site visit should take place to which all of the tenderers are invited. It is in the interests of both owner and Contracting organisations that the books are opened to provide maximum detail, keeping in mind that all of the tenderers are already bound by a non-disclosure agreement.

The owner's tender board should have decided upon the principles of their tender evaluation. The covering letter or questionnaire should reflect the intended evaluation categories to ensure that all of the required information is provided. The form of contract should be specified and the contractors should also be given the opportunity to propose alternatives or to take exception to the documents. An example of evaluation categories is presented below in the discussion of tender evaluation.

Tender Evaluation

A procedure, similar to that used for judging the pre-qualification submissions, is recommended for evaluating tenders.

The tender board will decide, before the bids are opened, on the individual categories that will comprise the evaluation matrix and the individual weighting of each category. Whereas, with the shortlisting evaluation, the submissions were judged on a "go", "no-go" basis, the tender evaluations will be scored according to the pre-determined criteria of the tender board.

An example for an engineering/construction project is presented below.

TABLE 2

Construction Management Tender Evaluation

Category			Weighting
1.	**TECHNICAL** (30% of Total Score)		
	(a)	**Personnel Proposed**	
		Process Engineering	7
		Cost Control	6
		Scheduling	4
		Contract Administration	3
		Construction Management	7
		Design Engineering	7
		Procurement	4
		Expediting and Inspection	3
	(b)	**Performance Record**	
		project Completion	6
		Industrial Relations	4
		Safety	7
		Quality Management	3
		Environment	5
2.	**PROCEDURES** (20% of Total Score)		
	(a)	**Reporting**	
		Costs	7
		Schedule	5
		Productivity	3
		Variances	7
		Presentation	4
	(b)	**Standards**	
		Engineering	7
		Procurement	4
		Inspection	2
		Safety	8
		Environment	7
3.	**TARGET SCHEDULE** (15% of Total Score)		
	Target Completion Date		10
	Proposed Bonus/Penalty Terms		4
	Mobilisation Schedule		8
4.	**TARGET PRICE** (35% of Total Score)		
	Target Price Proposal		10
	Proposed Bonus/Penalty Terms		4
	Exceptions Taken		10
	Exclusions		10

It is assumed that each member of the tender board will have had several days to read through the tender proposals. The procedure used in scoring the submission is to score in committee, ensuring that consensus is reached on each score. The best submission in each category rates a 10 score, and the other tenders are ranked according to relative merit. Generally the typical three to six tenders can be ranked in less than a day using this procedure, provided that the board members developed the categories and weightings. It is emphasised that the evaluation categorisation and relative weightings will vary considerably according to the owner project requirements.

Contract Negotiations

The final negotiations with the preferred tenderer or tenderers will comprise the contract clarification discussions and the final negotiation of commercial terms and conditions. These discussions are usually between the Owner Project Manager and Commercial Manager and their counterparts in the contractors' organisation.

The contract clarification discussion should be a line by line examination of the work scope, the proposed contract terms and conditions, to ensure that both parties have a similar appreciation of the project definition. It is worth emphasising that contracts are not needed in a perfect world. As we are committed to living in a less than perfect world, it is to the advantage of owner and contractor to minimise the contract imperfections prior to a commitment being made.

It is outside the scope of this paper to discuss the commercial side of contract negotiations. Each company will have its individual negotiating position. It is worth noting that the structured approach to definition and evaluation will provide an owner with a clearer vision of what they want from a contractor for their project. This, in itself, may provide the owner with a stronger negotiating position than would otherwise have been the case.

A finding of the Project Performance Predictability report noted an increase in contractor-client contractual disputes. This was partially ascribed to the limitations of A.S.2124.1986 resulting in owners and contractors developing their own contract terms and conditions. This variety of terms is creating concern in the industry and the Report recommends the development of uniform and equitable contract conditions.

CONTRACTOR MANAGEMENT

The final stage to ensuring the predictability of the cost performance is to ensure that the information flow to the owner during the implementation of the project is timely and contains the required information.

If the project is of sufficient dimension, e.g. engineering and construction of a major project, then the first stage to ensuring this information flow is to prepare a Project Procedures Manual. This manual will describe all the procedures that are required for the owner, contractor and sub-contractors to conduct the business required for the effective execution of the project. Interfaces with parties, such as regulatory authorities, may also be described. It will generally contain a description of:

1. Contractor and owner organisations.
2. Responsibilities and levels of authority.
3. Reporting requirements and procedures.
4. Cost and schedule control procedures.
5. Procedures for handling variances and scope changes.
6. Accounting and payment procedures.
7. Purchasing procedures.
8. Sub-contracting procedures.
9. Examples of standard contract forms and documents.
10 Reference of project standards and specifications.
11 .Miscellaneous procedures (Safety, First Aid, Environment, Industrial Relations, Quality Control, etc.).

It is a problem as to when this document should be prepared and by whom. It needs to be substantially in place before a final commitment to a contractor is made and, ideally, that contractor should be a party to its preparation. If either the owner or contractor has a standard procedures manual that can easily be modified for a particular purpose, then the preparation of the Project Procedures Manual would be one of the selected contractor's first tasks after commitment.

Cost and Schedule Control

The amount of detailed information required by the owner in the case of a lump sum or turnkey contract is far less than would be the case for other forms of contract. Because the information flow and the owner involvement is less for these cases, then the owner's project team would only comprise those personnel required to verify payments and to monitor, authorise and control variations.

Where the risk of cost and schedule overruns is primarily to the owner's account, the owner has to be more involved in the day-to-day progress and cost of the work. The owner would structure the composition of the project team accordingly, which would generally comprise engineering and site personnel plus cost control and accounting functions.

Generally project progress has been monitored on the basis of the accumulated monthly costs measured against the predicted S-curve of cashflow or value of work done. Cashflow is a poor indicator as, due to invoice and payment delays, it often represents the project status of two months previously. Value of work done (VOWD), provided it is measured consistently, is a better tool. A typical inconsistency relates to major equipment purchases. Often the VOWD measurement ignores equipment purchases until they arrive on site. Hence problems with cost and schedule are not indicated in time for remedial action to be taken.

Therefore cost curve monitoring is insufficient to track trends and variances and some form of physical progress measurement is required. Comparison between physical progress and VOWD can then be used to indicate project cost trends.

Physical progress can be tracked against the project bar chart and milestones. For example, in the case of a milling circuit, if the concrete pour has to be completed five days before mill installation commences, then project site staff will track the physical progress of the concrete, while off-site expediting staff will be tracking the manufacture and transport schedule of the mill to ensure conformity with the schedule. Installation hours actually expended against budgeted hours for any activity can be used as a good indication of likely cost and can signal potential variation claims or cost overruns.

It is the owner's responsibility to define reporting requirements and to make sure that these are carried out accurately and to the extent necessary to satisfy owner requirements. Only if potential problems are identified at the earliest opportunity, can the

appropriate action be taken. Only if owners and contractors fully understand project objectives and utilise the highest quality project management techniques can we hope to improve the predictability of project performance.

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